YAAD KAMRUL BARI

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Portfolio Yaad Bari Yaad Bari

CAREER OBJECTIVE

Computer Science graduate (CGPA 3.96/4.0, BRAC University) with hands-on experience in agentic AI, computer vision, and robotics. Led the AI and Machine Vision sub-team at BRACU Duburi, Bangladesh's first autonomous underwater vehicle, taking the perception pipeline from data curation to TensorRT deployment on a Jetson within a ROS2 stack. Builds LLM agents, multi-agent systems, and RAG applications with LangGraph, FastAPI, and PostgreSQL/pgvector. Seeking to join research-driven teams building reliable AI and autonomous systems for real-world, resource-constrained environments.

EXPERIENCE

Research Assistant **May 2026 – October 2026**
Dubotech Digital Ltd. *Dhaka, Bangladesh*
- Supported a confidential government study on capability development for large networked systems (under NDA).

Sub-Team Lead, AI and Machine Vision **April 2026 – July 2026**
BRACU Duburi, The First Autonomous Underwater Vehicle of Bangladesh *Dhaka, Bangladesh*
- Lead end-to-end ML pipeline for underwater object detection and tracking using YOLO and tracking algorithms, covering dataset curation, annotation, training, and deployment.
- Contributed to the team's 8th-place finish among 58 international teams at RoboSub 2026.

Undergraduate Teaching Assistant **February 2025 – May 2026**
Department of Computer Science and Engineering, BRAC University *Dhaka, Bangladesh*
- Conducted 15 hours of weekly consultation, supporting students with coursework concepts and problem-solving.
- Guided students through core algorithmic concepts, including algorithm design and analysis.

EDUCATION

BRAC University **September 2022 – June 2026**
Bachelor of Science in Computer Science, CGPA: 3.96/4.0 *Dhaka, Bangladesh*

Notre Dame College **July 2019 – December 2021**
Higher Secondary Certificate 2021, GPA: 5.0 *Dhaka, Bangladesh*

Dhaka Residential Model College **January 2011 – March 2019**
Secondary School Certificate 2019, GPA: 5.0 *Dhaka, Bangladesh*

PROJECTS

AUV Hull Inspection: Autonomous Ship-Hull Survey and Defect Detection | *MuJoCo, PyTorch, EKF, Docker*
- AUV fleet that surveys a ship's hull in simulation, detects corrosion, cracks and fouling, and recovers from faults.
- On 8 held-out hulls: 100% coverage with no false claims and 86% of defects found at 98% precision.

Agentic AUV: Fault-Tolerant Multi-Agent AI for an Autonomous Underwater Vehicle | *LangGraph, VLM*
- Multi-agent LLM system that plans underwater missions from plain English with tool calling and AI guardrails.
- Uses a VLM for scene understanding and multimodal sensor fusion (EKF, sonar) to keep working through sensor faults.

AUV Vision Pipeline | *YOLO11, PyTorch, TensorRT, ROS2, Jetson*
- Built the computer vision pipeline for Bangladesh's first AUV, from annotation to YOLO11 training (0.995 mAP50).
- Pruned the model and deployed it with TensorRT on a Jetson Orin Nano for real-time detection and ByteTrack tracking.

Camera–Sonar Perception: Underwater Detection and Fusion Tracking | *YOLO, Imaging Sonar, Tracking*
- Fuses YOLO camera detections with forward-looking sonar for multi-object tracking with Hungarian assignment.
- Completed 10/10 murky-water runs against 0/10 without tracking; fused position error ∼6 cm vs. ∼29 cm camera-only.

AUV Navigation Fusion: Underwater State Estimation and Sensor Recovery | *EKF, GTSAM/iSAM2, GRU*
- 8-state EKF fusing IMU, DVL, depth, compass and landmarks, benchmarked against a GTSAM/iSAM2 factor graph.
- Handles DVL dropouts with a GRU velocity model and sonar wall ranges, which cut error from 2.07 m to 0.03 m.

MuJoCo Marine Sim: Underwater Vehicle Dynamics and Fault Injection | *MuJoCo, Python, pytest*
- Underwater vehicle simulator with six-thruster dynamics, buoyancy and currents, plus camera, sonar, DVL and IMU.
- Injects 11 scheduled fault types (camera blackout, DVL dropout, leaks) and generates YOLO labels automatically.

AUV LLM Mission Planner: Agentic AI for Autonomous Underwater Vehicles | *LLM Agents, Ollama, ROS2*
- LLM agent that turns plain-English commands into robot missions using tool calling and structured outputs.
- AI guardrails validate every plan and the agent re-plans on failure, evaluated on 25 missions in Gazebo.

Weather-Robust Object Detection Using GANs | *PyTorch, Pix2Pix, YOLOv5*
- Developed a weather-aware object detection system for autonomous vehicles, using Pix2Pix GAN to generate synthetic adverse weather scenarios (snow, rain, fog).
- Trained YOLOv5 on the generated data for real-time detection that stays reliable in extreme weather.

bari: AI Housing Assistant for Students in Dhaka | *LangGraph, RAG, FastAPI, React, PostgreSQL/pgvector*
- Agentic AI assistant for finding safe housing in English and Bangla, using RAG, hybrid vector search and re-ranking.
- LangGraph background agent with human-in-the-loop approval; guardrails blocked 100% of prompt-injection attacks.

Pinstagram | *FastAPI, React Native, Next.js, PostGIS, pgvector*
- Instagram-style app for discovering places in Bangladesh, with a map, reviews, and bus and ride-hail directions.
- Added semantic search, RAG review summaries and an AI trip agent; async FastAPI backend with 163 integration tests.

AWARDS

International Competitions

- **8th Place, RoboSub 2026, RoboNation, Woollett Aquatics Center, California** – Secured 8th place overall among 58 international teams, contributing as Sub-Team Lead of AI and Machine Vision.
- **8th Place, RoboSub 2025, RoboNation, Woollett Aquatics Center, California** – Secured 8th place overall among 58 international teams, contributing as Technical Coordinator.

National Competitions

- **2nd Runner Up, National Telco Warfare Datathon 2024, East West University** – Achieved 2nd Runner Up in a national-level ML and DL competition, developing a VGG19-based image classification model with ∼90% accuracy.

TECHNICAL SKILLS

Programming: Python, C, C++, Java
AI & LLMs: Agentic AI, LLM Agents, Multi-Agent Systems, RAG, LangGraph, Tool Calling, VLMs, LLM Evaluation
Machine Learning: PyTorch, TensorFlow, OpenCV, YOLO, NumPy, Pandas, Matplotlib, Seaborn
Robotics: ROS2, Gazebo, MuJoCo, Pixhawk, Sensor Fusion, Kalman Filters
Backend & Databases: FastAPI, SQLAlchemy, REST APIs, PostgreSQL, PostGIS, pgvector, MySQL, Redis
Frontend: React, React Native, Next.js
Edge AI: Jetson Nano/Orin Nano, TensorRT, ONNX, Model Pruning, Edge Deployment
Tools: Docker, Git, GitHub Actions, pytest, Ollama
Soft Skills: Problem-Solving, Communication, Adaptability, Time Management, Integrity

REFERENCES

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